

Decision Theory and Social Choice

PE3101P — Lecture 1

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National University of Singapore

13th August, 2026

Today

- 1 Logistics
- 2 Course Overview
- 3 Two senses of "ought"
- 4 Acts, states, outcomes
- 5 Chances, credences, utilities
- 6 Bayesianism

Hello

- ▶ I'm your instructor for **PE3101P: Decision Theory and Social Choice**.

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- ▶ Please call me **Tomi**.

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- ▶ Please call me **Tomi**.
- ▶ No need for "Dr. Francis" or "Prof. Francis" or anything like that.

The register

- ▶ The University wants attendance records.

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- ▶ **Please sign the register** as it comes round.

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- ▶ Add your name and signature. In future we'll have printed copies.

Finding me

► **Email:** tomi.francis@nus.edu.sg

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- ▶ **Office hours:** Thursdays, 4–6pm, **AS7-03-12**.

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- ▶ (My office may move later this semester. I'll tell you if it does.)

What this course is about

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- ▶ (1) is **decision theory**.
 - ▶ (2) is **social choice theory**.

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Roughly half the semester on each. Decision theory first.

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- ▶ Trouble getting at anything? Email me.

The syllabus

- ▶ A printed copy should be in front of you — if not, grab one from the front.

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- ▶ Also on my website. (And on Canvas when I can, but the website is likelier to be current.)
- ▶ **Logistical question? Reread the syllabus first.** It may well answer it.

The textbook

Schwarz, *Belief, Desire, and Rational Choice* (2022)

Open access. A PDF is on my website.

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- ▶ Listed in the syllabus and on the reading list — see my website.

Questions and discussion

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- ▶ Tutorials are more discussion-focused — I'll want more from you there.

Use of AI

Assessed work

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Everything else

Use it or don't — but use your judgement. Help your learning, don't inhibit it.

Use of AI: good ideas

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- ▶ Triaging extra readings you've found, to see if they're worth your time.

Use of AI: bad ideas

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In short

Do use AI to supplement your learning. **Do not** use it to replace your learning.

We'll start with tables like this

	It rains	It doesn't rain
Take the umbrella	Dry, but carrying it	Dry, but carrying it
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- ▶ We'll learn to pick out the **states**, the **acts** and the **outcomes**.
- ▶ Then **dominance principles** — and why we must be careful with them.

Decision theory: what's coming

- **Probabilities** for states of nature: what they mean, what rules they obey, and why it's irrational to break those rules.

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- ▶ **Probabilities** for states of nature: what they mean, what rules they obey, and why it's irrational to break those rules.
- ▶ **Newcomb's problem**: what to do when you can **expect** things to go badly if you perform a certain act, even though you **know** performing that act will make things better.

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- ▶ (If that sounds impossible — stick around.)

Decision theory: the centrepiece

Expected Utility Theory

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- ▶ i.e. as though you'd put numbers on how good each outcome is, and were making the average as high as possible.
- ▶ Then: **money pump arguments** — if you don't choose that way, you can be made worse off for no reason, **and you can see it coming**.

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- ▶ **Sen's paradox**: even a minimal commitment to making people better off collides with liberal rights.
- ▶ **Arrow**: the impossibility of a satisfactory voting system.
- ▶ **Is voting irrational?** You almost certainly won't change the result.

So — let's start on decision theory.

The Miners Problem

THE MINERS PROBLEM

Ten miners are trapped in a shaft, threatened by rising water. You don't know whether they are in shaft *A* or shaft *B*.

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Ten miners are trapped in a shaft, threatened by rising water. You don't know whether they are in shaft *A* or shaft *B*. You have enough sandbags to block **one** shaft, not both. Block the right shaft and all survive; block the wrong one and all die. Do nothing, and both shafts half-fill: **one miner dies**.

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	Miners in A	Miners in B
Block A	all 10 live	all 10 die
Block B	all 10 die	all 10 live
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What should you do?

Three questions

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- ▶ What if the miners *are* in shaft A ? What then?

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- ▶ Should you block A , block B , or neither?
- ▶ What if the miners *are* in shaft A ? What then?
- ▶ What if the miners are in shaft A **and you still don't know it?**

Two senses of “ought”

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What you ought to do is whatever will *in fact* be best. You may have no idea what that is. Never mind.

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Evidence-relative

What you ought to do is whatever is best *given your evidence*. Here, what you know matters a lot.

Which one is this course about?

Decision theory is about the evidence-relative sense.

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How can we decide well, on the basis of the evidence we actually have?

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The **columns** are the **states** — or *states of nature*.

- ▶ Ways the world might be that you're **uncertain** about.
- ▶ That you **can't change**.
- ▶ That **determine** what each act leads to.

Reading the table

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The **cells** are the **outcomes** — or *final outcomes*, or *possible worlds*.

- ▶ A complete description of a possible result of your decision.

A question

	Miners in A	Miners in B
Block A	all 10 live	all 10 die
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Is the result of **Block neither** when the miners are in A
the same outcome as when they're in B ?

Suppose we had a fourth option

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- ▶ vs **Block A** : better if they're in B , same if in A .

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- ▶ vs **Block A** : better if they're in B , same if in A .
- ▶ vs **Block B** : better if they're in A , same if in B .
- ▶ vs **Block neither**: better **whichever** state obtains.

Two kinds of dominance

Weak statewise dominance

Definitely at least as good, and **possibly** better.

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So **Block both** strongly statewise-dominates **Block neither**.

So should we always choose dominant acts?

It seems so.

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It seems so.

Why accept a risk — or a certainty — of a worse outcome?

A counterexample?

A student wonders whether to study for an exam.

	Will Pass	Won't Pass
Study	Pass & No Fun	Fail & No Fun
Don't Study	Pass & Fun	Fail & Fun

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Does “Don’t study” statewise dominate “Study”?

Yes — and that's the problem

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Remember

States of nature are things you **can't change**.

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States of nature are things you **can't change**.

But studying *does* make a difference to whether you pass.

The fix

- ▶ We should choose statewise-dominant acts — **but only once the states are set up properly.**

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- ▶ (We'll see what that means later.)

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What we should do depends not just on the states and outcomes, but on **how likely** each state is.

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- ▶ If the world were very likely to end — have the party.
- ▶ In the actual world — Scrabble wins.

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- ▶ The probabilities we care about are **credences**, or *degrees of belief*.
- ▶ Roughly: how likely **you think** something is, given your evidence.
- ▶ When I say “probability”, I mean **credence**, unless I say otherwise.

Credences aren't objective chances

JACK AND JILL

Jack has flipped a coin and is looking at it. It landed heads.

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Jack has flipped a coin and is looking at it. It landed heads. Jill didn't see it, and is trying to guess.

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- ▶ The **chance** of tails is **zero**. It landed heads.
- ▶ Jill's **credence** in tails should be **50%**.

We care about credences because we often don't know the chances — and we want to decide well on the evidence we actually have.

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\$100 if a fair coin lands heads, \$0 if tails. The **expected** amount is **\$50**.

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Example

\$100 if a fair coin lands heads, \$0 if tails. The **expected** amount is **\$50**.

Note: the expected amount is \$50 even though **I know for certain I won't get \$50**.

Maximising expected utility

	Heads ($p = 0.5$)	Tails ($p = 0.5$)
Take the bet	\$100	\$0
Decline	\$50	\$50

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- Each row is a set of possible outcomes with probabilities attached.

It depends how you assign the utilities

Risk-neutral: every dollar worth the same

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Bet: **40**. Decline: **50**. **Don't bet.**

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Bet: **40.** Decline: **50.** **Don't bet.**

Both are ways of maximising expected utility.

One thing I haven't done

I haven't given you **any argument** that you *should* maximise expected utility.

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That comes in several weeks' time.

Which credences should we have?

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The Principle of Conditionalisation

They must change in response to evidence by **conditionalisation**.

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The axioms

- (i) $\text{Cr}(\sigma) \geq 0$
- (ii) $\text{Cr}(X) = 1$
- (iii) If σ and τ are **disjoint**, then $\text{Cr}(\sigma \cup \tau) = \text{Cr}(\sigma) + \text{Cr}(\tau)$

Next week: conditional probability

A sneak peek:

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Conditional probability

$$\text{Cr}(A \mid B) = \frac{\text{Cr}(A \cap B)}{\text{Cr}(B)} \quad \text{provided } \text{Cr}(B) > 0.$$

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See you next week.